

JEE MAIN REVISION MASTER: METALLURGY

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 10 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Isolation Of Elements — Core Principles & Governing Laws	5 PYQs	Core Principle
Moderate	Isolation Of Elements — Applied Cases, Reactions & Traps	5 PYQs	Applied Topics
Standard	Isolation Of Elements — Standard Calculations & Problem Archetypes	0 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Isolation Of Elements — Core Principles & Governing Laws	Fe_2O_3 , SO_2
Isolation Of Elements — Applied Cases, Reactions & Traps	$2\text{Al}_2\text{O}_3 + 3\text{C} \xrightarrow{\text{long}} 4\text{Al} + 3\text{CO}_2$, ΔG° , $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Isolation Of Elements — Core Principles & Governing Laws (Moderate Yield — 5 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Isolation Of Elements.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Isolation Of Elements.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Isolation Of Elements PYQs.

Representative Source PYQs:

- [2020 Shift] Q#1 (MCQ): "Function of potassium ethyl xanthate in froth floatation process is to make the ore..." [Ans: N/A]
- [2020 Shift] Q#2 (MCQ): "Sulphide ore on roasting gives a gas X. X reacts with Cl_2 in the presence of activated charcoal to give Y. Y is..." [Ans: N/A]
- [2020 Shift] Q#4 (MCQ): "Copper is extracted from copper pyrites by..." [Ans: N/A]

3.2 Isolation Of Elements — Applied Cases, Reactions & Traps (Moderate Yield — 5 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Isolation Of Elements.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "The incorrect statement about Hall -Heroult process is..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "Select the correct statement :..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "For the formation of which compound in Ellingham diagram ΔG° becomes more and more negative with increase in temperature?..." [Ans: A]

3.3 Isolation Of Elements — Standard Calculations & Problem Archetypes (Standard Yield — 0 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Isolation Of Elements.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "The incorrect statement about Hall -Heroult process is..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "Select the correct statement :..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#1)

The incorrect statement about Hall -Heroult process is

- (A) carbon anode is oxidised to CO and CO₂ .
- (B) Na₃AlF₆ helps to decrease the melting point of the electrolyte.
- (C) CaF₂ helps to increase the conductivity of the electrolyte.
- (D) oxidation state of oxygen changes in the overall cell reaction.

Correct Answer: None

Solution Analysis: The incorrect statement about Hall-Heroult process is given in option (d). It is because in this process oxidation state of oxygen remains same in the overall cell reaction. $2\text{Al}_2\text{O}_3 + 3\text{C} \rightarrow 4\text{Al} + 3\text{CO}_2$

Benchmark Problem 2: KCET 2020 (Q#1)

Function of potassium ethyl xanthate in froth floatation process is to make the ore

- (A) lighter
- (B) hydrophobic
- (C) hydrophilic
- (D) heavier

Correct Answer: None

Solution Analysis: The role/function of potassium ethyl xanthate in the froth floatation process is to increase the non-wettability of mineral particles by making it hydrophobic. These are also given the name collectors.